

Day 3 – Data Visualization

Today is important because eventually he will need to **look at sensor patterns** to understand why a motor might be failing.

Learn

Study:

- What is data visualization?
- Why graphs are useful
- Matplotlib
- Line graphs
- Bar charts
- Histograms
- Labels
- Titles
- Legends

Practical Work

Use the motor dataset.

Program 1 – Temperature Over Time

Create a line graph showing:

Time → Temperature

The graph should have:

- X-axis: Time
- Y-axis: Temperature
- Title: Motor Temperature Over Time

Program 2 – RPM Over Time

Create a graph showing:

Time → RPM

Program 3 – Current Over Time

Create:

Time → Current

Program 4 – Compare Two Sensors

Create a graph comparing two measurements.

For example:

Temperature
Current

over time.

Think About the Data

After creating your graphs, answer:

1. Does the temperature remain stable?
2. Are there sudden increases?
3. Does RPM remain stable?
4. Does current change significantly?
5. Do any variables appear to change together?

Write your answers in a short text file:

observations.txt

Git Task

Commit and push today's work.

Day 4 – Statistics & Detecting Abnormal Values

Now we're getting closer to his actual final-year project.

Learn

Study:

- Mean
- Median
- Minimum
- Maximum
- Range
- Standard deviation
- Why averages alone can sometimes be misleading

Use Pandas to calculate these values.

Practical Work

Program 1 – Motor Statistics

Calculate:

Temperature

- Average
- Minimum
- Maximum
- Standard deviation

Voltage

- Average
- Minimum
- Maximum

Current

- Average
- Minimum
- Maximum

RPM

- Average
- Minimum
- Maximum

Display something like:

MOTOR STATISTICS

Temperature
Average: 61.4
Minimum: 55.2
Maximum: 78.1

RPM
Average: 1482.4
Minimum: 1430
Maximum: 1510

Program 2 – Identify Abnormal Readings

Create rules for identifying potentially abnormal readings.

For example:

Temperature > 75 → HIGH TEMPERATURE

Current > 10 → HIGH CURRENT

RPM < 1400 → LOW RPM

The program should identify records that violate these thresholds.

Example:

WARNING

Time: 09:42
Temperature: 78.4
Status: HIGH TEMPERATURE

Program 3 – Motor Health Status

Create a simple health classification:

NORMAL
WARNING
CRITICAL

For example:

NORMAL
→ All sensor readings are within expected range.

WARNING
→ One sensor is outside the normal range.

CRITICAL
→ Multiple sensors are outside the normal range.

The exact thresholds can be chosen by him for this exercise.

Important: He should document why he chose those thresholds rather than treating them as scientifically correct motor limits.

Git Task

Commit and push today's work.